

Kotipalli Pranamesh Sri Durga Deepak

+91 95021 20678 | pranamesh67@gmail.com | GitHub | LinkedIn | Rajahmundry, Andhra Pradesh

Summary

Computer Science undergraduate skilled in Java, Python, and Spring Boot with strong fundamentals in DSA, OOP, and DBMS. Experienced in backend development, REST APIs, RAG/LLM systems, and data-driven applications.

Education

Bachelor of Technology in Computer Science

Expected May 2027

Vellore Institute of Technology, Vellore, Tamil Nadu

CGPA: 8.32/10

Higher Secondary Education

May 2023

Tirumala Junior College, Andhra Pradesh

Class XII: 87%

Secondary School Education

May 2021

Tirumala E.M. High School, Andhra Pradesh

Class X: 100%

Skills

- **Languages:** Java, Python, SQL, C, C++, JavaScript, TypeScript
- **Frameworks & Backend:** Spring Boot, Spring Security, REST APIs, JWT Authentication
- **Databases & Tools:** MySQL, PostgreSQL, MongoDB, Docker, AWS, Git, GitHub
- **Generative AI:** Retrieval-Augmented Generation (RAG), LangChain4j, Gemini API, Vector Databases, Vector Search, Prompt Engineering
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Projects

InfraSub – Multimodal InSAR Subsidence Digital Twin

May 2026 – Aug 2026

Technologies: Java 17, Spring Boot 3, PostgreSQL, PostGIS, pgvector, LangChain4j, React, Mapbox GL JS

- **Engineered** a Java 17 / Spring Boot 3 backend with Spring Security/JWT, leveraging **PostGIS** geospatial queries to map millimeter-level InSAR ground deformation to infrastructure assets.
- **Developed** a multimodal fusion API using **LangChain4j** and **pgvector** to combine spatial telemetry with engineering-document retrieval, generating explainable, asset-specific risk diagnostics through LLM reasoning.

Enterprise Document Management and Q&A System

2025

Technologies: Java, Spring Boot, RAG, Vector Database, Gemini API

- **Developed** an enterprise document Q&A system using RAG to answer user questions from uploaded PDF documents.
- **Implemented** PDF text extraction, chunking, local embeddings, and vector similarity search using Qdrant.
- **Integrated** LangChain4j with the Gemini API to generate answers from retrieved document context with source page references.

FinFlow Dashboard

2025

Technologies: TypeScript, Zustand, Recharts, CSS

- **Developed** a responsive finance dashboard for monitoring balances, income, expenses, spending categories, and transactions.
- **Implemented** transaction search, filtering, sorting, CSV export, and role-based Admin/Viewer access.
- **Built** interactive financial visualizations using Recharts with centralized state management via Zustand.

Certifications

- **AWS Cloud Practitioner Essentials** – June 2026
- **Generative AI Using IBM Watsonx** – June 2025

Achievements

- Secured **Top 5** position at the **Technex Hackathon, IIT BHU Varanasi**.
- Solved **100+ Data Structures and Algorithms problems** on LeetCode and GeeksforGeeks.
- **NSS Club & RnR Coordinator, VIT Vellore** – Coordinated community service initiatives, contributed to a 7-day NSS camp, and managed registration and reception operations for Rivera 2026.